Hailong Qu

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Research Interests in AI for Science

Wind and structural engineering Turbine power and fatigue loads, material bond strength, and structural capacity.
Building and atmospheric environments Indoor air quality and atmospheric monitoring.
Medical imaging Fracture recognition and localization.
Engineering agents Engineering software use, 3D navigation, and object manipulation.

Education

Chongqing University, School of Civil Engineering Ph.D. student in Civil Engineering 2025–present
Dalian University of Technology Master's degree in Civil Engineering 2022–2025
China University of Mining and Technology Bachelor's degree in Civil Engineering 2018–2022

Internship

Zhiman AI Engineering Agent R&D Intern **June 2026 – November 2026**

- **EngiWorld benchmark** Co-first author (listed second). Helped design engineering file checks and ran the model evaluations. Seven models were tested on a benchmark of 1,301 tasks, six fields and 26 applications. Overall execution success on cross-software collaboration tasks was 3.6%.
- **AI-assisted engineering software development** Designed and tested a Micius-Flash case study. A model built a CAD plug-in from requirements and host documentation. Verified hole positions and reports for linear and circular patterns in the host software. [Technical report]
- **3D modeling software** Replayed the same actions in Blender and a PySide6 prototype over 10 build-and-test iterations. Verified primitive creation, mesh editing, 3D viewport controls, and material assignment.

Publications, Preprints and Accepted Manuscripts 2026 Xinrui rankings

Wind Energy and Wind Field Reconstruction

[1] Predicting power and fatigue loads of yawed large offshore wind turbine based on cross-scale transfer learning
The 5 MW to 15 MW transfer matched XGBoost's accuracy with 40% of target simulations versus 80% for XGBoost.
Ocean Engineering **(Tier 1, TOP)** [DOI] · First author · Published (2026)

Building Environment and Indoor Air Quality

[2] Probabilistic Short-Term Forecasting of Spatially Heterogeneous Indoor CO_2 Fields with Graph-Spectral Increment Diffusion
Used graph-spectral increment diffusion to forecast the distribution of indoor CO_2 concentrations across locations.
Building and Environment **(Tier 1, TOP)** · First author · Accepted

Structural Engineering and Composite Materials

[3] A machine learning model based on GAN-ANN data augmentation for predicting the bond strength of FRP-reinforced concrete under high-temperature conditions
Augmented high-temperature pull-out data with a GAN, compared six regressors, and interpreted predictions with SHAP.
Composite Structures **(Tier 1, TOP)** [DOI] · First author · Published (2025)

[4] Prestressed UHPC Beam Flexural Capacity Prediction Improvement with Constraint-Guided Neural Networks
Used reinforcement ratio and material strength to constrain flexural capacity predictions for prestressed UHPC beams.
International Journal of Bridge Engineering, Management and Research [DOI] · First author · Published (2025)

Engineering Agents and Multimodal Spatial Reasoning

[5] EngiWorld: What Can Frontier Agents Deliver in Professional Engineering Environments?
Evaluated agent-generated engineering files against geometric, physical, and design constraints in professional software.
arXiv [Paper] [Code] [Project site] · Co-first author (listed second) · Public preprint (2026)

[6] SpatialWorld: Benchmarking Interactive Spatial Reasoning of Multimodal Agents in Real-World Tasks
Evaluated first-person navigation and object manipulation across 760 tasks in eight simulated environments.
arXiv [Paper] [Code] · Co-first author (listed second) · Public preprint (2026)

Technical Skills

Programming and evaluation Python, GUI/CLI experiments, automated checks, trace replay, and failure analysis.
Engineering modeling Cross-scale transfer, GAN augmentation, constrained networks, XGBoost, and SHAP attribution.